

Lesson Cards

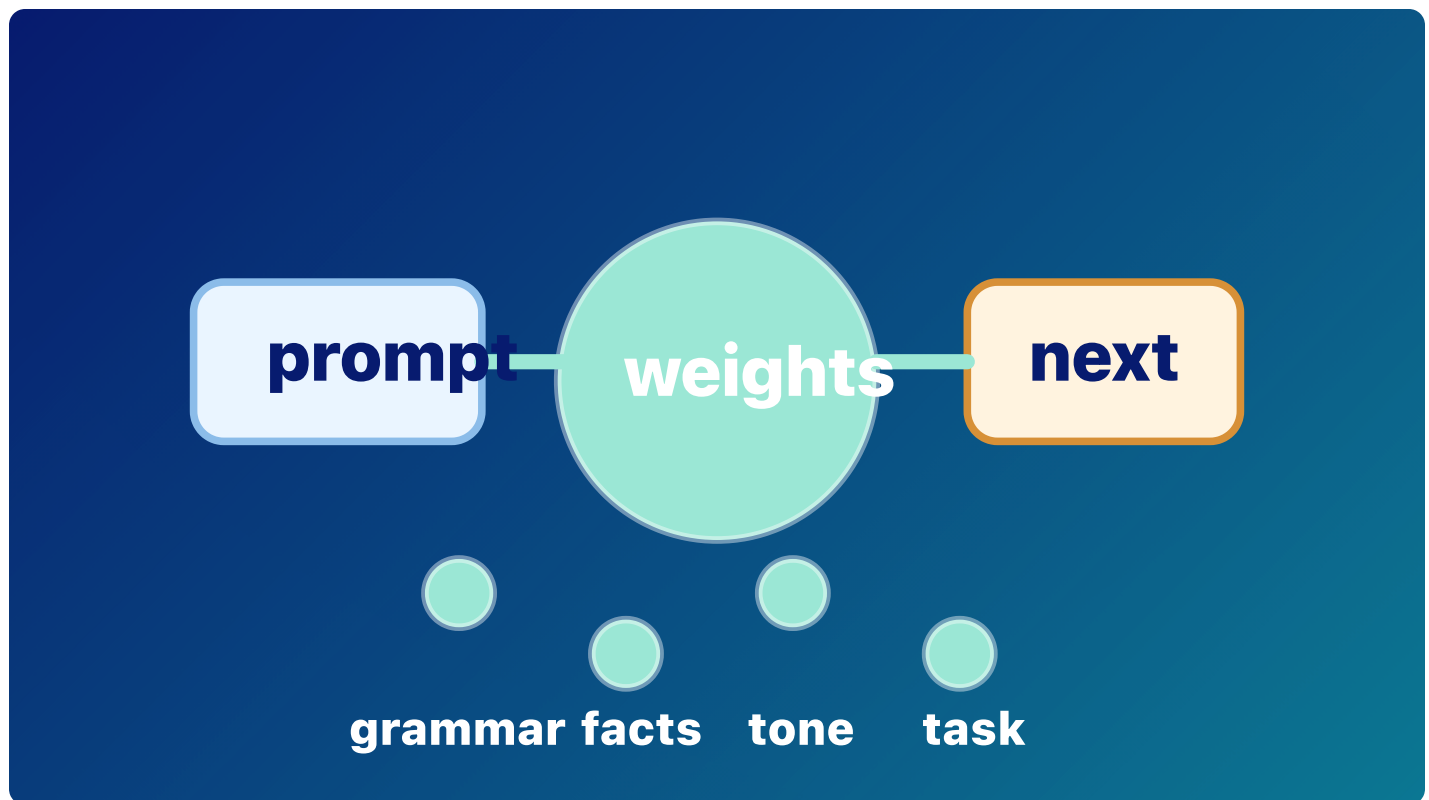
One accessible review card per lesson, built from the same HTML lesson content used in the app.

BEFORE MORNING

1 / 25

What Is an LLM?

A learned next-token prediction system



Prompt to Prediction

A prompt enters the current context; learned weights shape next-token probabilities.

DEFINITION

A large language model is a learned system that predicts likely next tokens from the context it can see.

WHERE

It sits between the current input context and the next response token.

WHY

This definition lowers the mystery: the model is powerful because training shaped many weights, not because it has a hidden mind.

RELATIONSHIP

During inference, the current context enters a forward pass, one generated response token is chosen, and that token is appended before the next pass.

METAPHOR

A very large autocomplete engine with learned structure, not a mind or a database.

BRAIN BRIDGE

Like a brain, many small signals can combine into useful patterns.

BRAIN LIMIT

Unlike a brain, an LLM has no lived body, feelings, personal goals, or awareness.

CHECKPOINT

Which description is most accurate? Answer: A learned prediction system.

prompt

response

token

weight

inference

Two AI Traditions

Rules versus learned patterns

weights
durable

context
temporary

retrieval
open book

different changes, different risks

Rules and Learned Patterns

Symbolic AI follows explicit rules; deep learning uses learned parameters.

DEFINITION

Symbolic AI manipulates explicit rules and symbols; deep learning adjusts many weights until useful patterns are captured.

WHERE

LLMs mainly come from the deep-learning tradition, even when they imitate rule-like reasoning in language.

WHY

The distinction explains why LLMs can be flexible and fluent without being hand-programmed rule by rule.

RELATIONSHIP

A prompt is not matched against one fixed rulebook; it is processed by learned parameters that were shaped across many examples.

METAPHOR

A rulebook versus a weather system of learned relationships.

BRAIN BRIDGE

The learned-pattern tradition is loosely like practice shaping future behavior.

BRAIN LIMIT

The model is not practicing with intention; training is an optimization process over data.

CHECKPOINT

Which phrase best describes the LLM tradition? Answer: A learned prediction machine.

pretraining

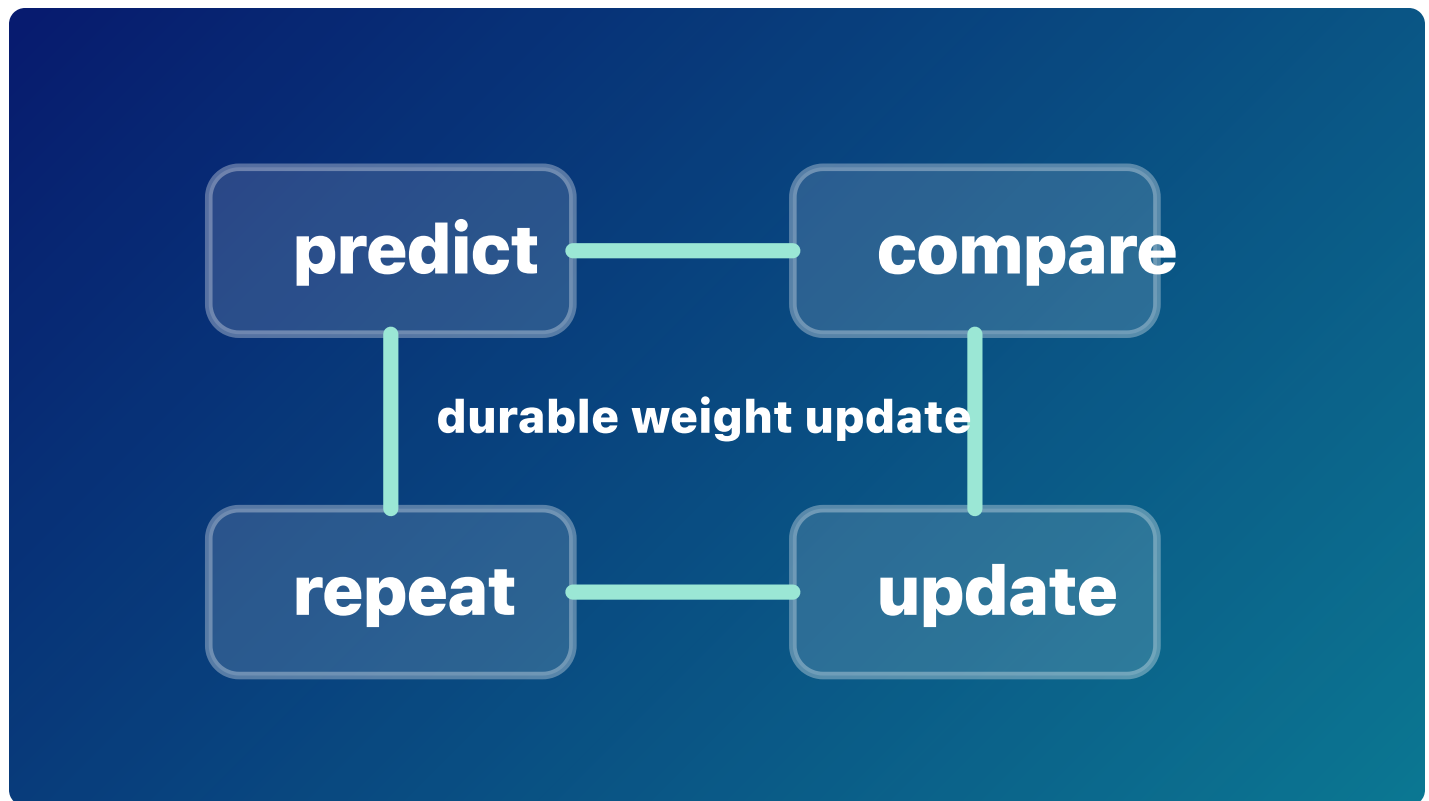
training

symbolic AI

deep learning

Training

Prediction error updates weights



Training Loop

Predict, compare, update weights, repeat.

DEFINITION

Training uses examples, predicts targets such as next tokens, measures error, and durably updates model weights.

WHERE

It happens before or outside ordinary use, in a separate optimization process.

WHY

Training is where durable capability is installed into the model.

RELATIONSHIP

Training changes parameters; inference uses those parameters to generate response tokens without normally rewriting them.

METAPHOR

Practicing a piece until the instrument settings are tuned before the performance.

BRAIN BRIDGE

It is fair to say repeated experience shapes later behavior.

BRAIN LIMIT

The model has no intention or memory of a lesson; weights are adjusted by math, not reflection.

CHECKPOINT

What makes training different from inference? Answer: Training durably updates weights.

training

weight

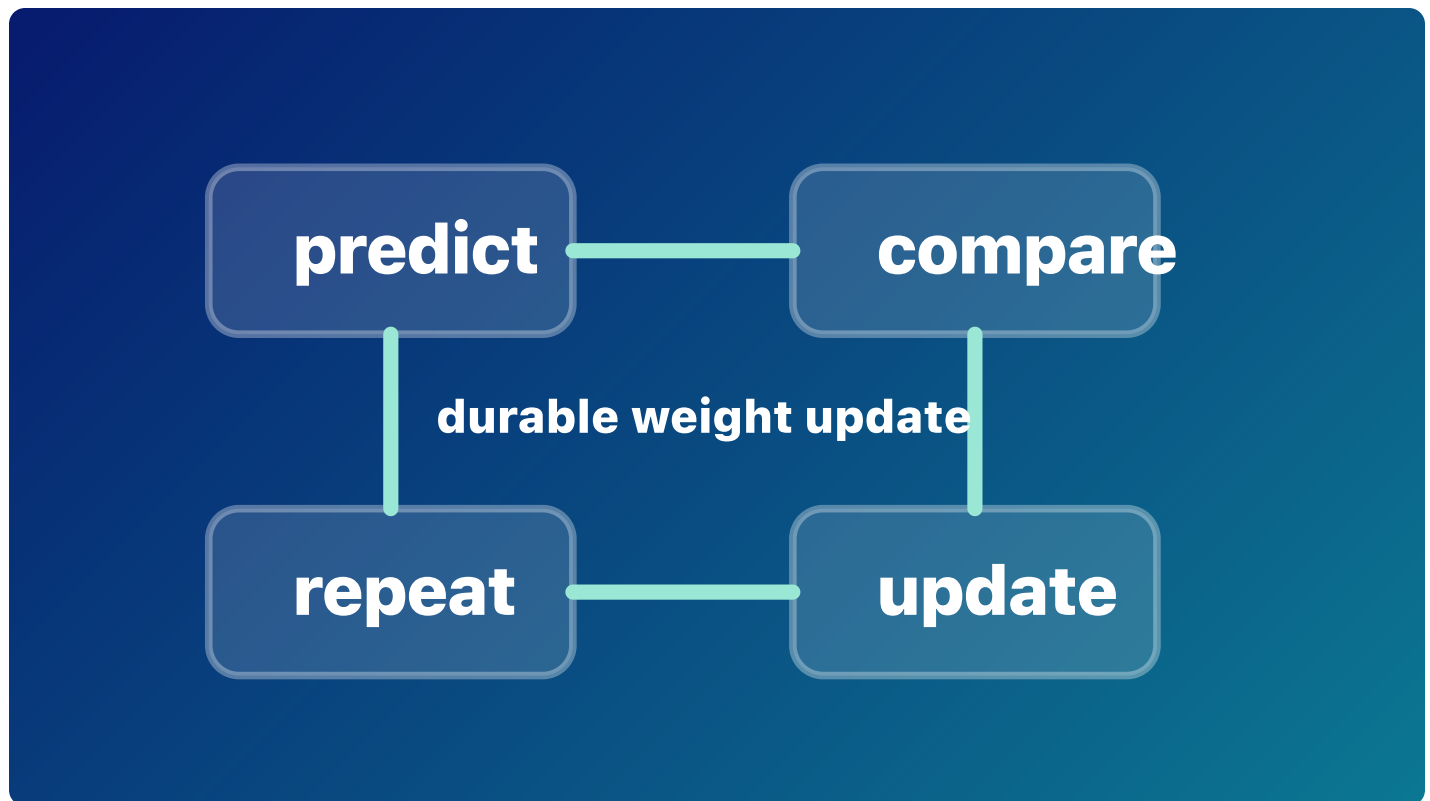
parameter

inference

forward pass

Pretraining

Broad durable learning before normal use



Broad Pretraining

Many prediction updates carve broad capability into model weights.

DEFINITION

Pretraining is broad early training that repeatedly predicts tokens across large datasets and durably updates model weights.

WHERE

It happens before a general model is released for everyday prompting.

WHY

Pretraining gives the model broad language patterns, factual associations, styles, and task structures.

RELATIONSHIP

Pretraining changes the model durably; inference only uses the weights that already exist.

METAPHOR

Billions of tiny nudges carving paths through a vast landscape.

BRAIN BRIDGE

It resembles broad education in the loose sense that later performance depends on prior exposure.

BRAIN LIMIT

The model does not understand lessons the way a person does; it stores learned statistical structure in parameters.

CHECKPOINT

During pretraining, do model weights change? Answer: Yes.

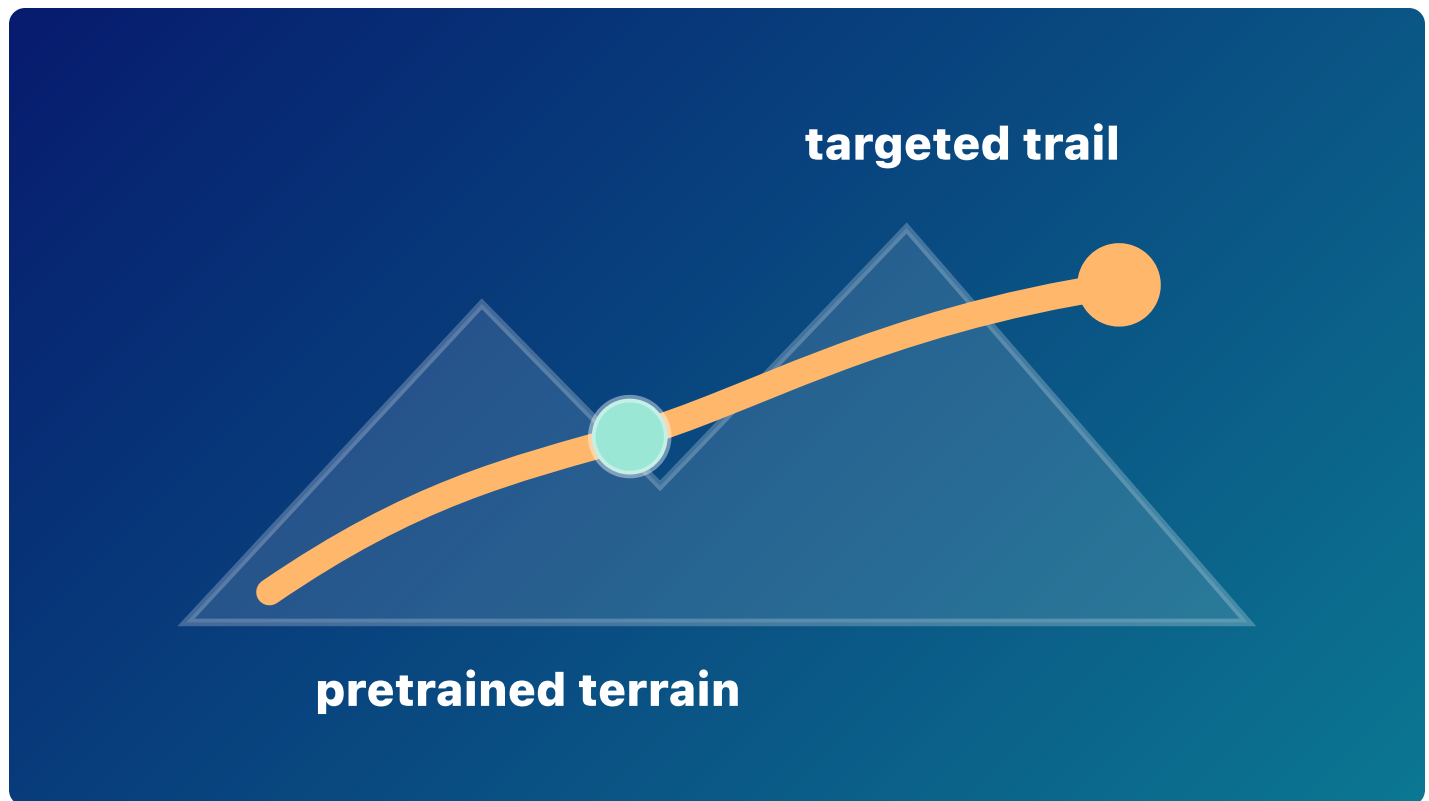
pretraining

training

inference

Fine-Tuning

Targeted durable shaping



Fine-Tuning Path

Targeted examples nudge an already-trained model toward a desired pattern.

DEFINITION

Fine-tuning is additional targeted training on examples, preferences, or domain data after broad pretraining.

WHERE

It happens after pretraining and before or between deployment cycles.

WHY

Fine-tuning can make future responses more helpful, specialized, or aligned with a task.

RELATIONSHIP

Pretraining builds broad capability; fine-tuning can durably change weights or add adapter weights so future responses follow a desired pattern.

METAPHOR

Carving useful trails through an already huge terrain.

BRAIN BRIDGE

It is a little like coaching a person toward a style of performance.

BRAIN LIMIT

The model is not adopting values or intentions; the output pattern is being optimized.

CHECKPOINT

During ordinary inference, do model weights durably update? Answer: No.

fine-tuning

pretraining

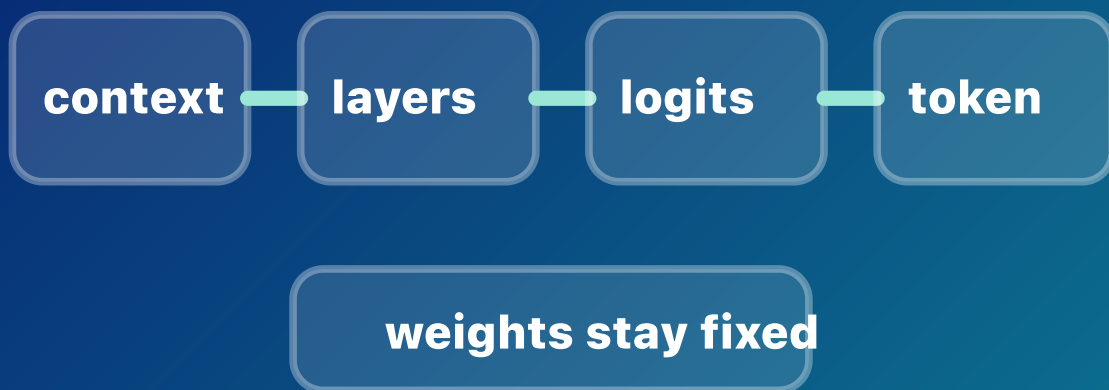
inference

prompt

response

Inference

A forward pass, not a memory update



Forward Pass

Inference creates temporary states while durable weights stay fixed.

DEFINITION

Inference is normal model use: fixed weights process the current context to generate response tokens without durable weight updates.

WHERE

It happens every time a deployed model answers a prompt.

WHY

This is the line between using a trained model and training a model.

RELATIONSHIP

Each forward pass turns the current context into next-token scores; decoding chooses one generated token for the response.

METAPHOR

Using a map to choose a route, not redrawing the map.

BRAIN BRIDGE

It can feel like thinking because internal representations change temporarily.

BRAIN LIMIT

Those hidden states are temporary computations, not memories, beliefs, or learning.

CHECKPOINT

What changes during ordinary inference? Answer: Temporary hidden states.

inference

forward pass

prompt

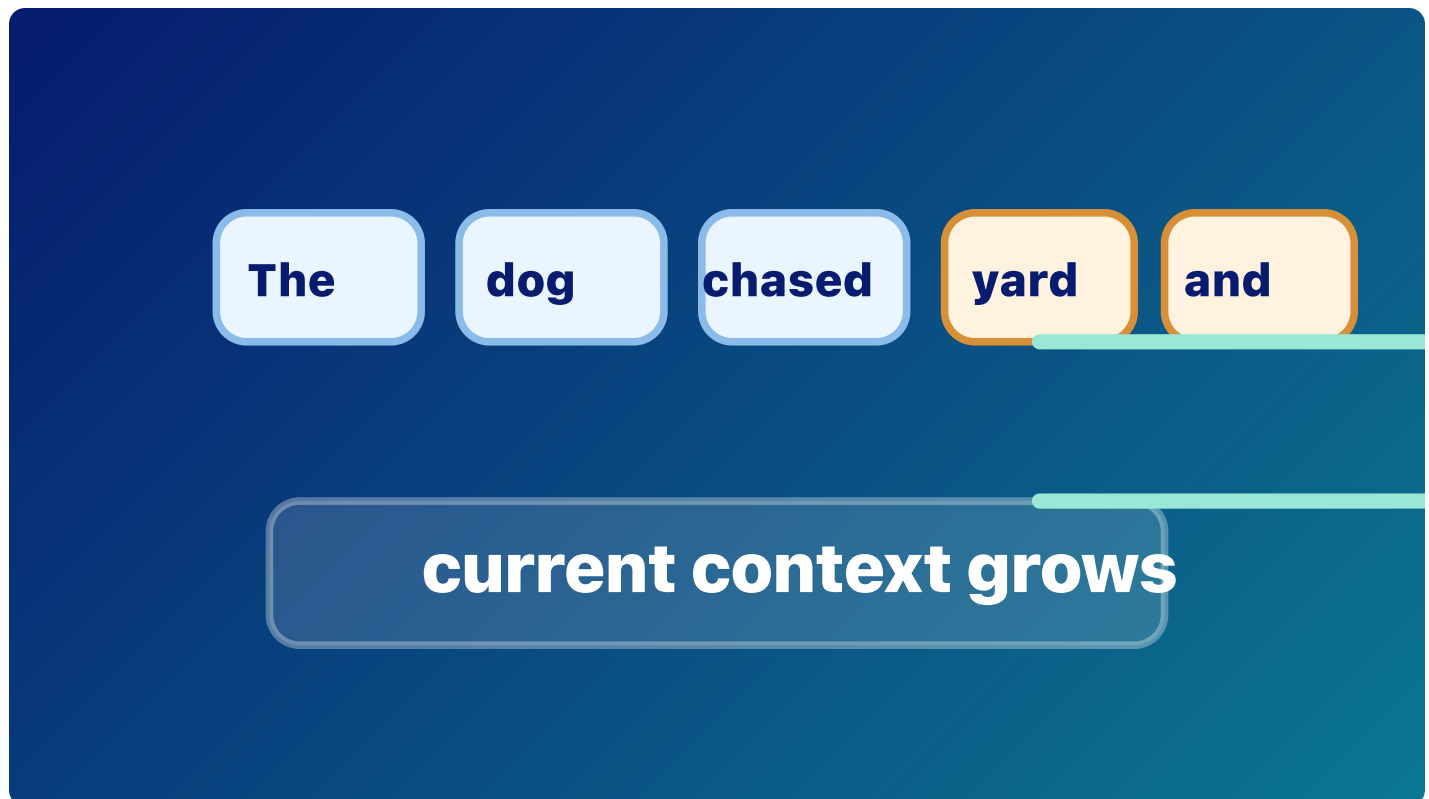
response

hidden state

logits

Prompt vs Response

Given context versus generated tokens



Prompt vs Response

Prompt tokens are given; response tokens are generated and appended.

DEFINITION

The prompt is the input context supplied to the model; the response is the sequence of tokens the model generates after reading it.

WHERE

Prompt tokens are present before the next token is chosen; response tokens are added one at a time as generation proceeds.

WHY

The distinction prevents a common myth that the model writes a full answer all at once.

RELATIONSHIP

Each generated response token is appended to the context, so it can influence the next forward pass.

METAPHOR

Given cards on a table, then new cards added one at a time.

BRAIN BRIDGE

It resembles a person continuing a sentence after reading a prompt.

BRAIN LIMIT

The model is not deciding what it means to say; it samples from learned next-token probabilities.

CHECKPOINT

What happens after one response token is generated? Answer: It is appended to the context.

prompt

response

prompt tokens

response tokens

generated token

context window

Tokenization

Text becomes chunks

Explain attention simply

Explain

attention

simply

Text to Tokens

Text is split into model-readable chunks before embedding lookup.

DEFINITION

Tokenization splits prompts and generated responses into chunks the model can represent as token IDs.

WHERE

It happens before embedding lookup and before the transformer layers process the input.

WHY

Models do not carry raw English sentences through their layers; they carry token IDs and numerical vectors.

RELATIONSHIP

Prompt tokens and earlier response tokens in the input context become token IDs, embeddings, tensors, and temporary hidden states.

METAPHOR

Parcels on a conveyor belt entering the model.

BRAIN BRIDGE

It loosely resembles hearing a sentence as parts before interpreting it.

BRAIN LIMIT

The parts are tokenizer chunks, not human phonemes, concepts, or meanings.

CHECKPOINT

What comes right after text is split into tokens? Answer: Token IDs and embedding lookup.

token

prompt tokens

response tokens

token-id

embedding

Token IDs

Chunks become lookup numbers



lookup key, not meaning

Token IDs

Each token gets a lookup number that points to an embedding row.

DEFINITION

A token ID is the number assigned to a token so the model can look up its learned starting vector.

WHERE

It sits between tokenization and the embedding table.

WHY

The ID is the bridge from text chunks to the learned numerical space the model uses.

RELATIONSHIP

Prompt tokens and already-generated response tokens use token IDs when they are part of the current input context.

METAPHOR

A library call number that points to the right shelf in the embedding table.

BRAIN BRIDGE

The lookup step is faintly like recognizing a familiar symbol.

BRAIN LIMIT

The number itself is not understanding; it is a key into a learned table.

CHECKPOINT

Why does the model need token IDs? Answer: To look up embeddings.

token-id

prompt tokens

response tokens

embedding

Embeddings

Token IDs look up learned vectors



many numerical features

Embedding Lookup

A token ID retrieves a learned starting vector.

DEFINITION

An embedding is a token ID's learned starting vector.

WHERE

It is retrieved from the embedding table at the start of model processing.

WHY

Embeddings give each token a learned numerical starting point before context reshapes it.

RELATIONSHIP

Prompt tokens and already-generated response tokens are looked up as embeddings when they are inside the current context.

METAPHOR

A token gets a starting address in a giant meaning cloud.

BRAIN BRIDGE

It loosely resembles a starting association a person might have with a word.

BRAIN LIMIT

An embedding is not a definition or memory; it is a learned vector row.

CHECKPOINT

What is an embedding in this app? Answer: A learned starting vector.

embedding

input context

token-id

vector

hidden state

Vectors

Lists of numbers carrying features



many numerical features

Feature Vector

A vector is a list of numerical features, not a sentence.

DEFINITION

A vector is a list of numbers that represents learned features of something, such as a token.

WHERE

Embeddings and hidden states are both vectors at different moments in the model path.

WHY

Vectors let the model compute with many fuzzy features at once instead of using one plain-English label.

RELATIONSHIP

An embedding vector starts the token; hidden-state vectors are later context-shaped versions of that token.

METAPHOR

A coordinate in a many-dimensional map.

BRAIN BRIDGE

It can resemble a bundle of associations around a word.

BRAIN LIMIT

The vector is not a conscious concept; it is a numerical representation useful for computation.

CHECKPOINT

Which statement is most accurate? Answer: Embeddings and hidden states are vectors.

vector

embedding

hidden state

feature

Tensors

Organized blocks of vectors



tokens by features by layers

Tensor Block

Tensors organize many token vectors for layer-by-layer processing.

DEFINITION

A tensor is an organized block of numbers, often holding many token vectors at once.

WHERE

Tensors carry the whole current context through transformer layers.

WHY

A tensor keeps the model's data organized by positions and features, so many tokens can be processed together.

RELATIONSHIP

Embeddings from the current context become tensors so the transformer can process prompt tokens and earlier response tokens together.

METAPHOR

A stack of spreadsheets carrying token features through the model.

BRAIN BRIDGE

It loosely resembles organizing many signals at once.

BRAIN LIMIT

The tensor is not a mental image or thought; it is a shaped numerical array.

CHECKPOINT

Which is the better definition of tensor here? Answer: An organized block of numbers.

tensor

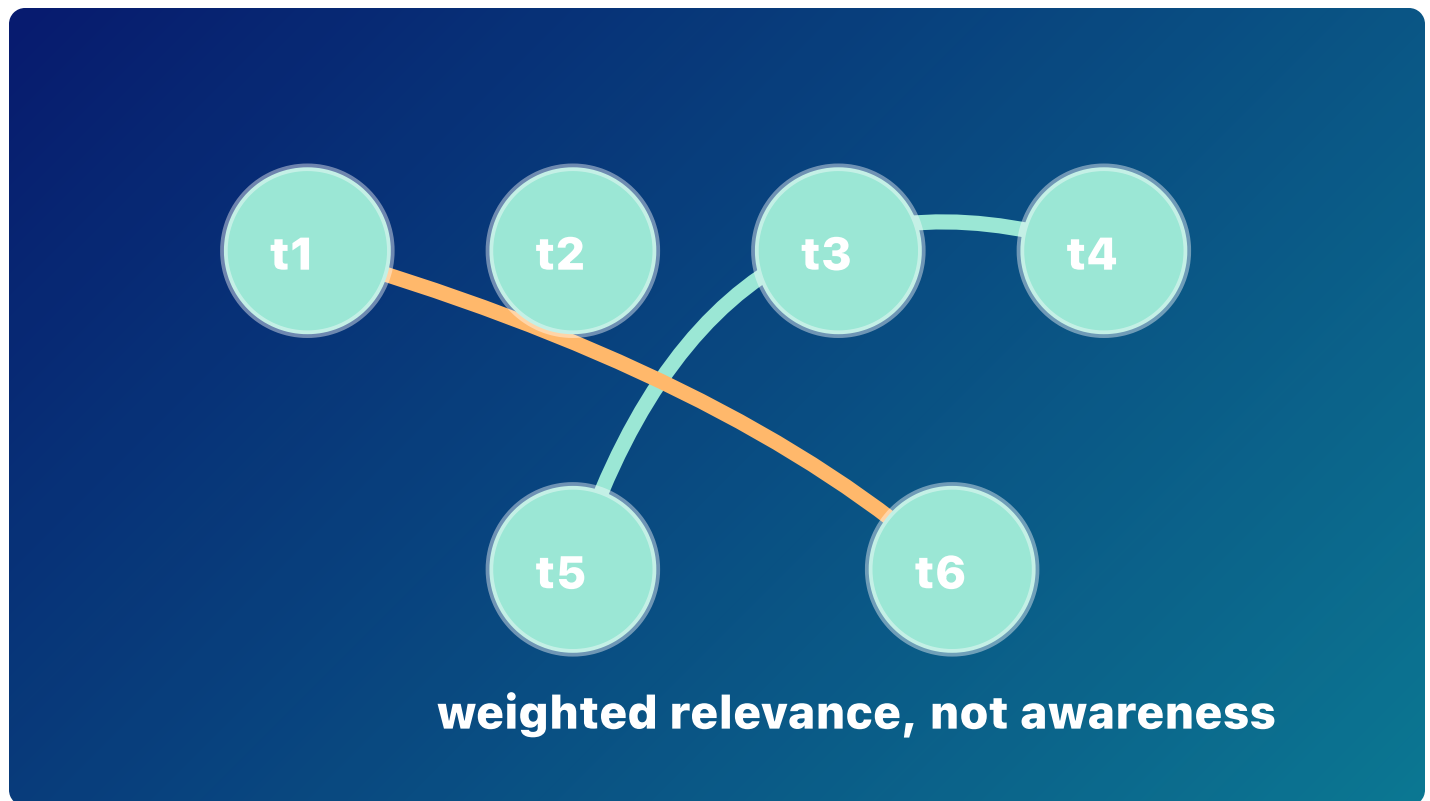
vector

input context

layer

Attention

Weighted relevance between positions



Attention Weave

Attention is weighted relevance between token positions.

DEFINITION

Attention assigns weighted relevance between token positions.

WHERE

It happens inside transformer layers during a forward pass.

WHY

Attention lets one token position use information from other visible token positions in the current context.

RELATIONSHIP

During inference, attention operates over token positions in the current context, including prompt tokens and earlier generated response tokens.

METAPHOR

Spotlights showing which earlier tokens are useful right now.

BRAIN BRIDGE

The word attention is useful because some parts of the context matter more than others.

BRAIN LIMIT

It is not human attention, awareness, desire, or focus. It is weighted relevance in a computation.

CHECKPOINT

Which tokens can attention look across during inference? Answer: Tokens currently in the context window.

attention

input context

prompt tokens

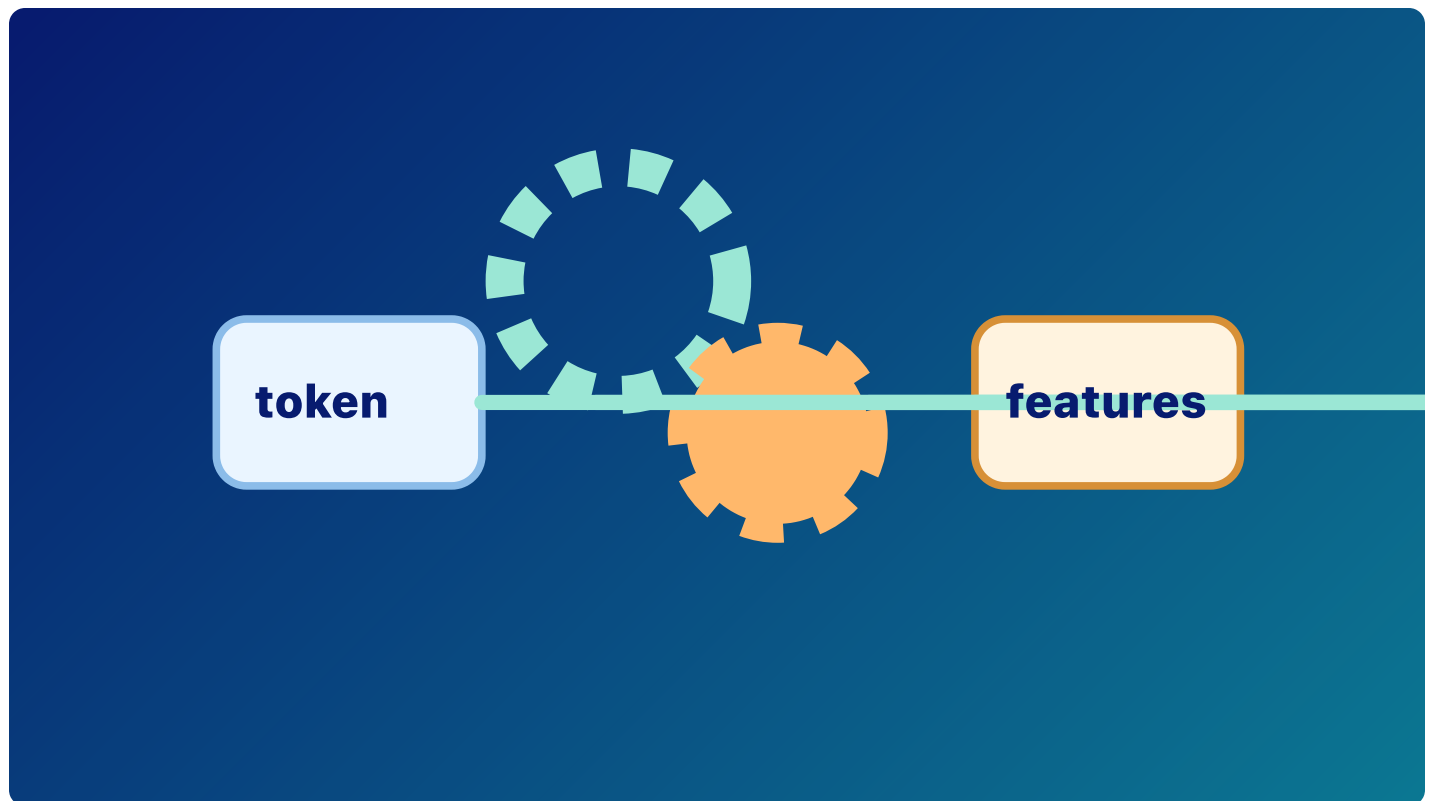
response tokens

hidden state

layer

MLP

Feature reshaping per token



MLP Reshape

The MLP reshapes each token position feature vector.

DEFINITION

An MLP is a feed-forward network that reshapes each token position's feature vector.

WHERE

It appears inside transformer layers, usually after attention has mixed information across positions.

WHY

The MLP helps turn mixed context into more useful per-token features for later layers.

RELATIONSHIP

During a forward pass over the current context, attention mixes information across positions and the MLP reshapes features within each position.

METAPHOR

A forge that bends each token vector into a more useful shape.

BRAIN BRIDGE

It is somewhat like a small processing step after noticing relevant context.

BRAIN LIMIT

The MLP is not a neuron with feelings or beliefs; it is a learned function applied to vectors.

CHECKPOINT

What does the MLP mainly do here? Answer: Reshapes per-token features.

MLP

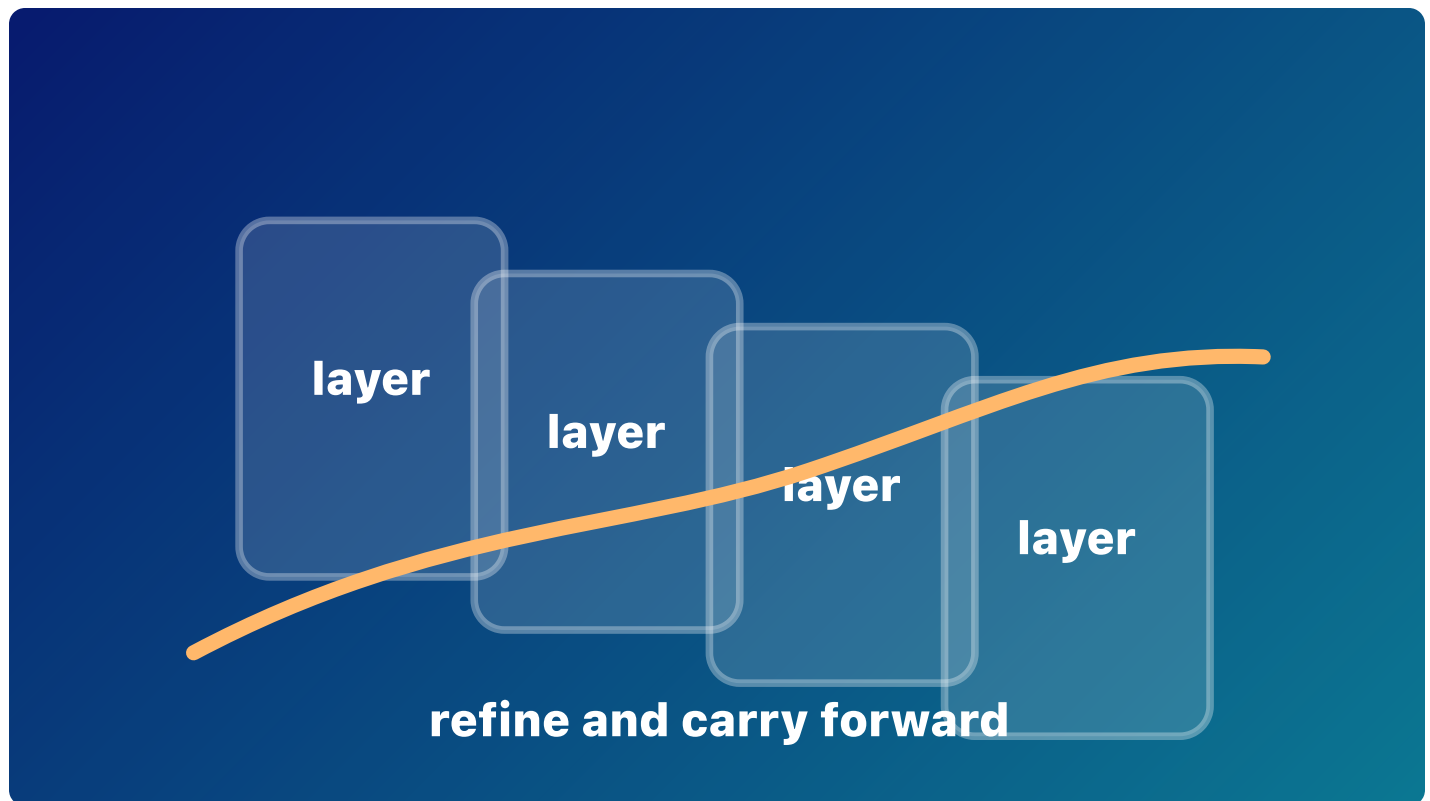
input context

hidden state

attention

Layers

Repeated refinement



Layer Stack

Repeated blocks refine hidden states while carrying useful signal forward.

DEFINITION

A transformer layer is a repeated block that usually combines attention, MLP feature reshaping, residual paths, and normalization.

WHERE

Layers are stacked inside the model between embeddings and next-token scores.

WHY

Repeated layers let simple starting vectors become richer, context-shaped hidden states.

RELATIONSHIP

Each layer refines hidden states while residual paths help useful information carry forward.

METAPHOR

A series of editing passes where each pass can revise but also preserve the draft.

BRAIN BRIDGE

The stack can feel like stages of processing.

BRAIN LIMIT

The layers do not form a human-like chain of thought; they update numerical states.

CHECKPOINT

What do layers repeatedly refine? Answer: Hidden states.

layer

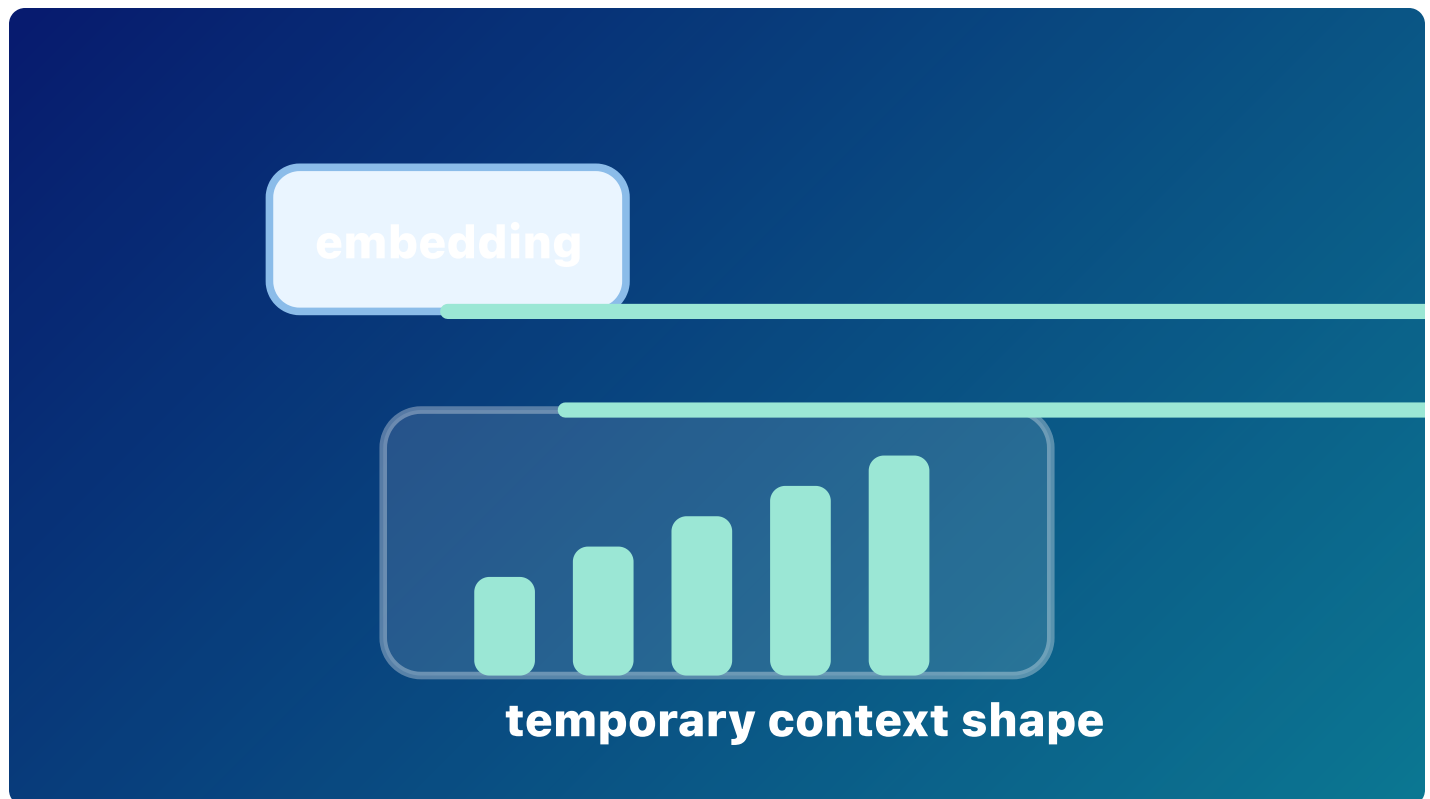
attention

MLP

hidden state

Hidden States

Temporary contextual vectors



Temporary Hidden State

Hidden states are context-shaped vectors created during a forward pass.

DEFINITION

A hidden state is the model's temporary internal context-shaped vector for a token at a given layer.

WHERE

Hidden states appear during the forward pass after embeddings enter the layer stack.

WHY

They are where the prompt's context changes the token representation before the model scores possible next tokens.

RELATIONSHIP

Hidden states are created while processing the current context during a forward pass; they are not the visible response.

METAPHOR

A scratchpad of numbers that changes while the prompt is being processed.

BRAIN BRIDGE

Hidden state can resemble temporary working memory.

BRAIN LIMIT

It is not permanent memory, consciousness, or private English text; it is a temporary vector.

CHECKPOINT

Why is it called hidden state? Answer: It is internal numbers, not visible text.

hidden state

input context

forward pass

embedding

layer

inference

Logits

Raw next-token scores

yard



street



banana



raw scores, not probabilities

Raw Scores

Logits are raw next-token scores before probabilities.

DEFINITION

Logits are raw scores for the next token before they become probabilities.

WHERE

They appear near the end of a forward pass, after the final hidden state is projected toward the vocabulary.

WHY

Logits show that the model has not chosen a token yet; it has scored candidates.

RELATIONSHIP

A forward pass over the current context produces logits for one next generated token.

METAPHOR

A scoreboard before the points are converted into odds.

BRAIN BRIDGE

It can feel like considering options.

BRAIN LIMIT

The model is not weighing options with intent; logits are raw numerical scores.

CHECKPOINT

Where do probabilities come from in this path? Answer: Softmax converts logits.

logits

generated token

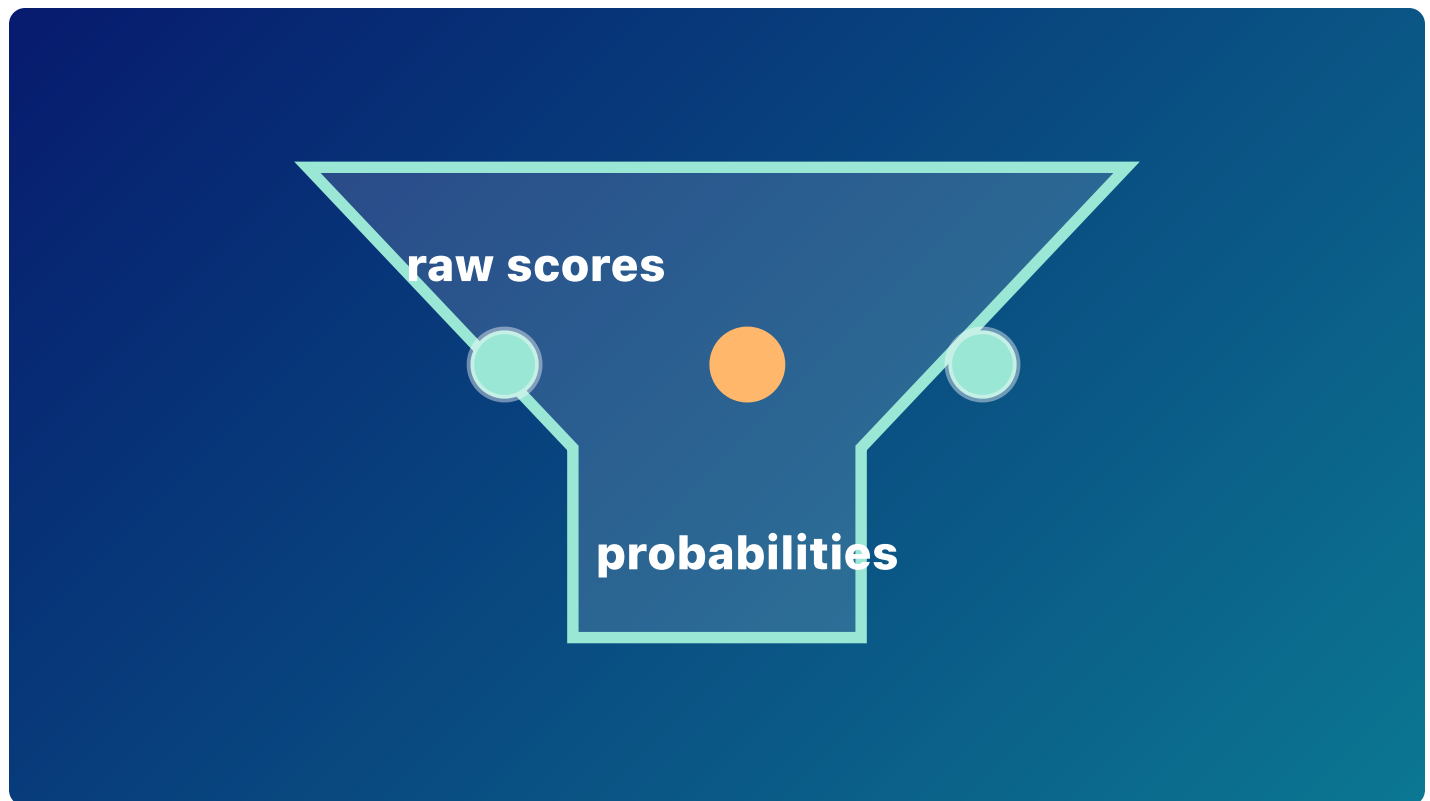
forward pass

hidden state

softmax

Softmax

Scores become probabilities



Softmax Funnel

Softmax converts raw scores into probabilities that sum to one.

DEFINITION

Softmax turns next-token raw scores into probabilities that sum to one.

WHERE

It sits between logits and sampling during a decoding step.

WHY

Softmax makes scores comparable as a probability cloud over candidate tokens.

RELATIONSHIP

Softmax converts logits into probabilities; sampling then chooses one token from those probabilities.

METAPHOR

A funnel turning scoreboard points into odds.

BRAIN BRIDGE

It resembles turning vague preferences into relative chances.

BRAIN LIMIT

The probabilities are mathematical outputs, not confidence, desire, or truth.

CHECKPOINT

What does softmax produce? Answer: Next-token probabilities.

softmax

logits

decoding step

sampling

probability

Sampling

One next token is chosen



Sample One Token

Sampling chooses one token from the probability cloud.

DEFINITION

Sampling chooses one next token for the response from the probability cloud.

WHERE

It happens after softmax in the decoding step.

WHY

Sampling explains why the same prompt can sometimes produce different good responses.

RELATIONSHIP

Temperature, top-k, and top-p shape the candidate pool before the chosen response token is appended to the context.

METAPHOR

Choosing from a weighted bowl of possible next tokens.

BRAIN BRIDGE

It can resemble choosing a word while speaking.

BRAIN LIMIT

The model is not choosing with intention; a decoding rule selects from probabilities.

CHECKPOINT

When the model chooses one token, what happens next? Answer: The token is appended and the model runs again.

sampling

generated token

decoding step

response tokens

temperature

top-k

top-p

Autoregression

One token, append, repeat



Append and Repeat

The chosen token is appended, then the model runs again.

DEFINITION

Autoregression means response generation by repeated next-token prediction: choose one token, append it, and run again.

WHERE

It is the loop that builds text during inference.

WHY

Autoregression explains why earlier generated tokens become part of what the model can use next.

RELATIONSHIP

Each generated response token becomes part of the input context for the next decoding step.

METAPHOR

A train adding one car at a time.

BRAIN BRIDGE

It resembles continuing a thought step by step.

BRAIN LIMIT

The model is not holding a plan in mind; it repeatedly predicts the next token from visible context.

CHECKPOINT

How does an LLM normally generate text? Answer: One token at a time.

autoregression

completion

generated token

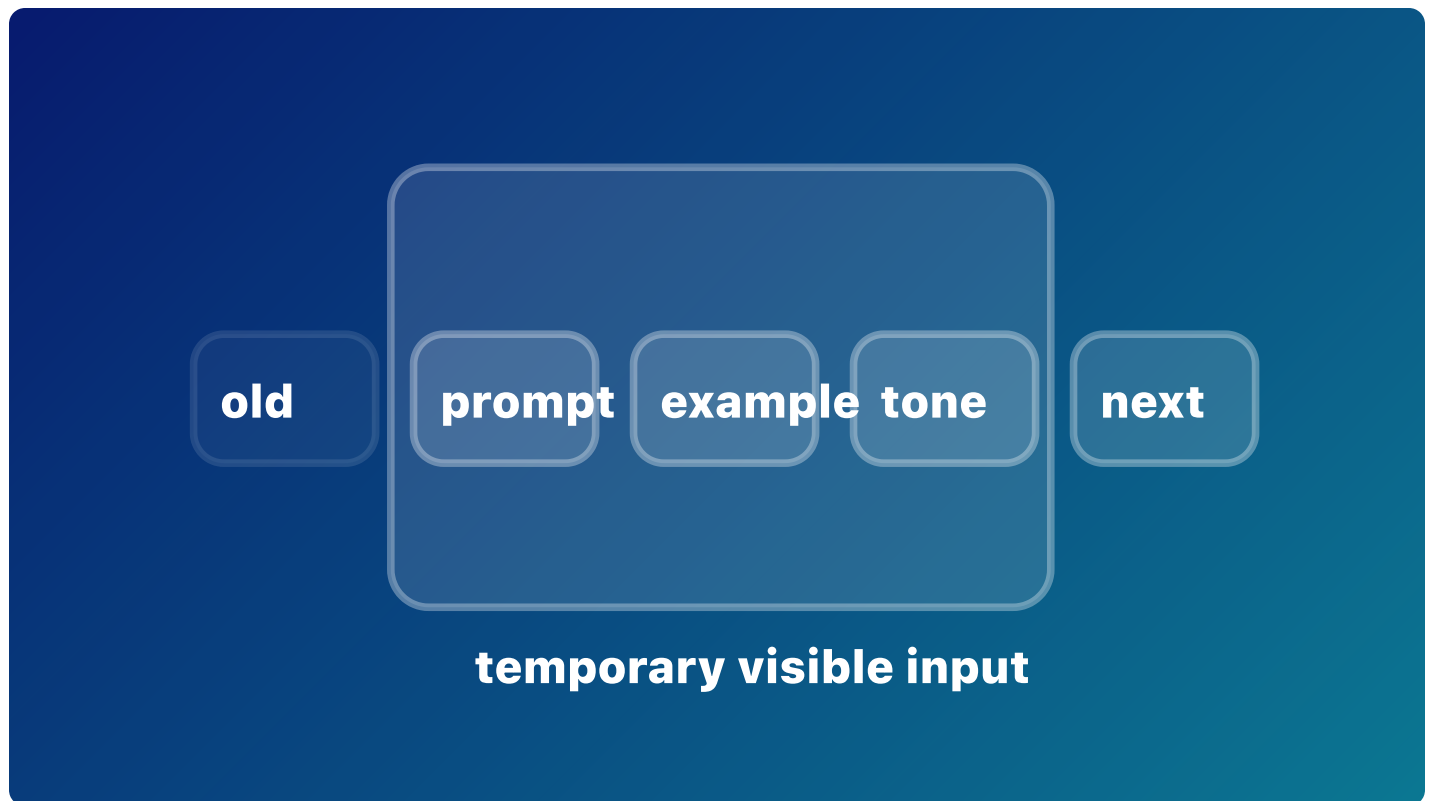
response tokens

sampling

inference

Context Window

Temporary working context



Temporary Context Window

Only visible context can influence the next token.

DEFINITION

A context window is the limited amount of tokens or media the model can currently use.

WHERE

It surrounds the prompt, conversation history, retrieved material, and response-so-far that remain visible.

WHY

It explains why models can use current context but do not automatically form permanent memory from every chat.

RELATIONSHIP

The model can only use tokens currently inside the context window, whether they came from the prompt, prior conversation, retrieved documents, or the response-so-far.

METAPHOR

A moving spotlight over a growing train.

BRAIN BRIDGE

It is loosely like working memory: what is currently available matters.

BRAIN LIMIT

It is not durable memory. What falls out cannot be directly attended to in the next pass.

CHECKPOINT

What can happen when the context window is full? Answer: Older tokens may be truncated.

context window

input context

prompt tokens

response tokens

rag

autoregression

inference

How AI Learns

Durable learning versus temporary steering



Learning Modes

Durable training, retrieval, and temporary steering change different things.

DEFINITION

AI systems can improve or behave differently through durable training, targeted fine-tuning, retrieval, or temporary in-context steering.

WHERE

These processes happen at different times: before deployment, between deployments, or inside one current run.

WHY

Calling every behavior change learning creates confusion about privacy, memory, and institutional risk.

RELATIONSHIP

Pretraining and fine-tuning can durably change weights; retrieval and examples in the prompt change only what the model can see during inference.

METAPHOR

Revising the textbook versus bringing notes into an open-book exam.

BRAIN BRIDGE

The word learning is useful when prior experience changes later behavior.

BRAIN LIMIT

In-context examples are not personal learning; they steer the current run without normally updating weights.

CHECKPOINT

Which process normally changes only the current context, not weights? Answer: Retrieval-augmented generation.

training

pretraining

fine-tuning

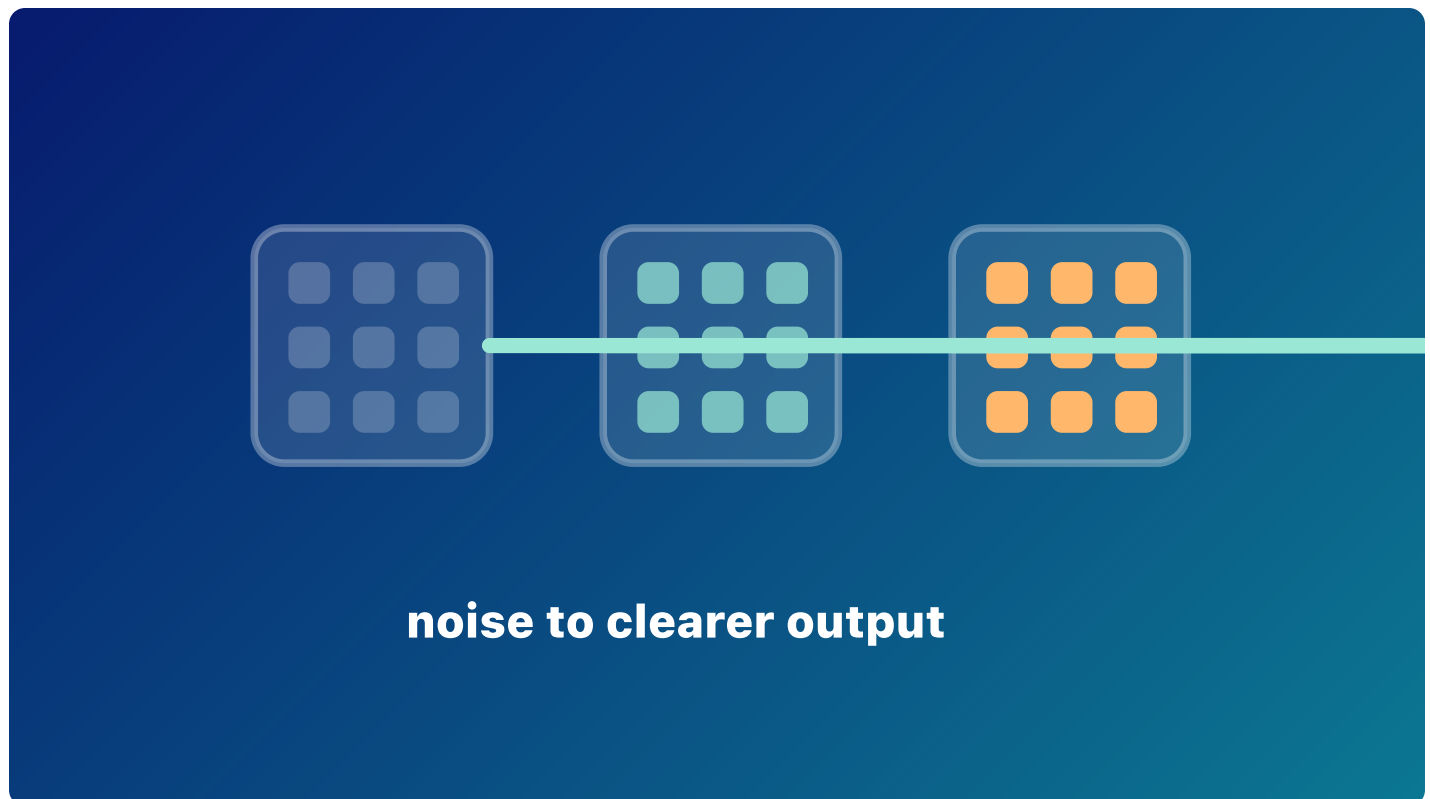
inference

rag

in-context learning

Diffusion vs Autoregression

Denoising is a different generation pattern



Denoise, Not Append

Diffusion refines noise step by step instead of generating text token by token.

DEFINITION

Diffusion generation usually starts from noise and denoises step by step.

WHERE

Diffusion is common in image generation and other media workflows, not the standard loop for text LLM generation.

WHY

The comparison prevents learners from treating all generative AI as the same mechanism.

RELATIONSHIP

Autoregressive LLMs build text token by token; diffusion models refine noise into an image or other output.

METAPHOR

Writing the next word versus developing a photograph out of static.

BRAIN BRIDGE

Both systems can feel imaginative because they produce new outputs.

BRAIN LIMIT

The mechanisms differ: diffusion denoises a representation; autoregressive text predicts and appends tokens.

CHECKPOINT

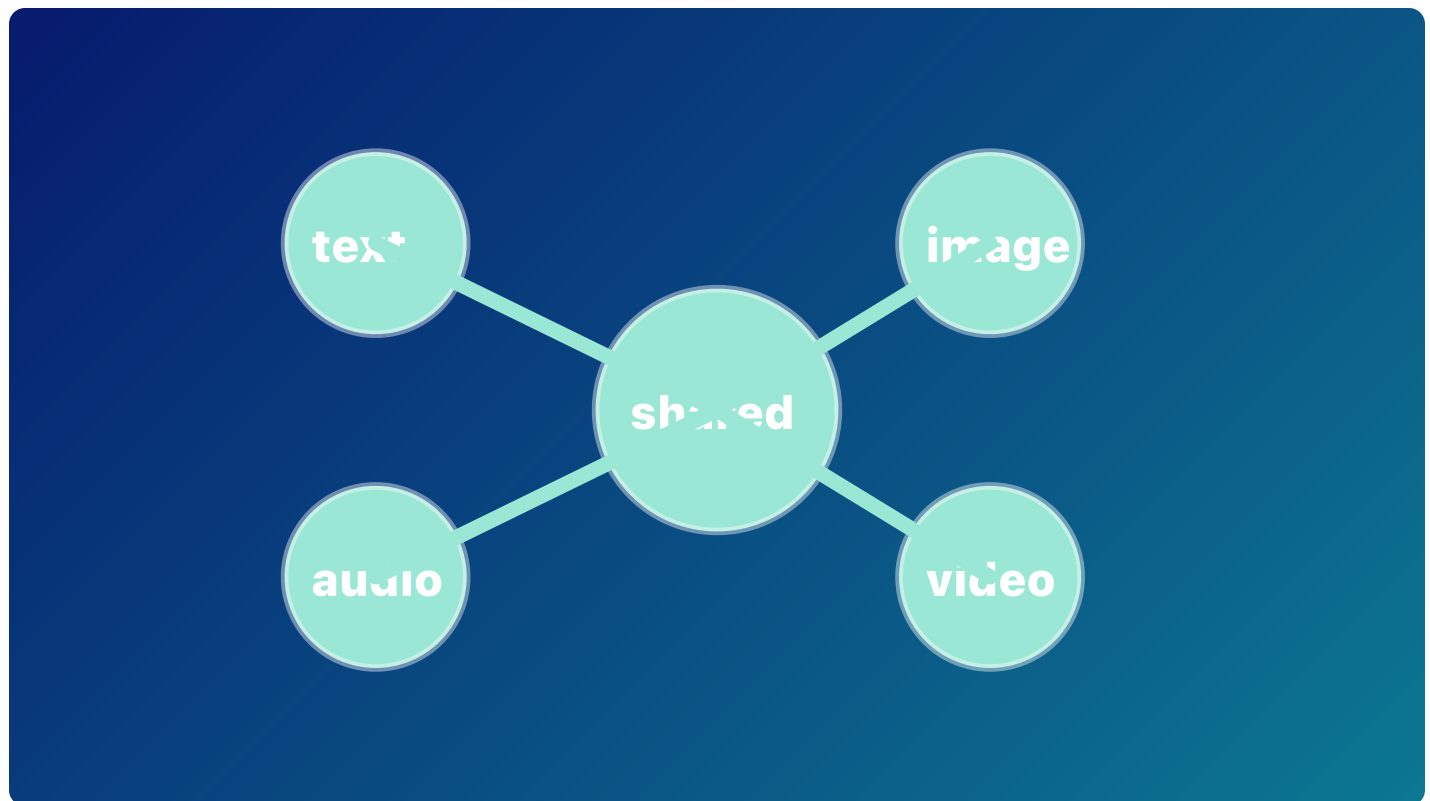
What does a diffusion model usually start with? Answer: Noise.

diffusion

autoregression

Multimodal AI

Multiple media types together



Shared Media Hub

Different media types can connect through learned representations.

DEFINITION

Multimodal AI can represent or process multiple media types, such as text, images, audio, or video.

WHERE

It appears when systems connect encoders, decoders, or shared representations across media types.

WHY

It explains how a system can read an image, answer in text, or combine speech and documents.

RELATIONSHIP

Different modalities may use connected encoders, shared vector spaces, or coordinated model components.

METAPHOR

A transit hub where different kinds of information can transfer lines.

BRAIN BRIDGE

People combine sight, speech, and text naturally, so the metaphor helps.

BRAIN LIMIT

Machine modalities are engineered representations, not human sensory experience.

CHECKPOINT

What does multimodal mean? Answer: Multiple media types.

multimodal

embedding

vector

Risk vs Myth

Practical institutional literacy



Risk or Myth

Clear mechanisms help separate practical risk from magical stories.

DEFINITION

Risk literacy separates practical harms from magical stories about what models are.

WHERE

It matters whenever people use AI in classrooms, research, administration, health, advising, or IT systems.

WHY

Clear mental models help institutions manage privacy, accuracy, bias, over-trust, and security without inventing magical threats.

RELATIONSHIP

Most real risks come from data, outputs, integrations, incentives, and human over-reliance; ordinary inference does not make the model secretly train itself.

METAPHOR

A campus safety map that marks real hazards clearly.

BRAIN BRIDGE

Brain metaphors can make risks feel urgent and memorable.

BRAIN LIMIT

Do not infer consciousness, intent, or secret self-training from fluent text. Keep the mechanism visible.

CHECKPOINT

Which is a real institutional risk? Answer: Uploading private data.

inference

training

context window

privacy

hallucination

prompt-injection